

Skill and Mental Fortitude: Factors behind Top Finishes on the PGA Tour

AM115 Mini-Project 1

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Abstract—In this project, we determine which factors contribute to Top-10 success on the PGA Tour with machine learning techniques. 2015-2018 will represent our train set, and we test on the 2019 season. First, we delineate which factors reflect a golfer’s skill and mental fortitude. After determining each feature’s relative importance, we extract each feature’s weight. Using a linear combination of these weights, we develop a new ranking system. Finally, we leverage the new system to predict whether a player will finish in the top 10% of all FedEx Cup Point earners with 85% accuracy.

I. INTRODUCTION

IN the middle of the 20th century, the chess world was experiencing an internal crisis. The Harkness system, which ranked chess masters against one another, lacked statistical rigor, and unfairly buoyed some players over others. To address this issue, Hungarian-American physics professor, Arpad Elo, stepped into the fold. The Elo System predicts a winner of a match by examining the difference in the ratings between two players. After a match, the ratings are updated, and over time, each player finds a rating equilibrium that accurately reflects their skill.¹

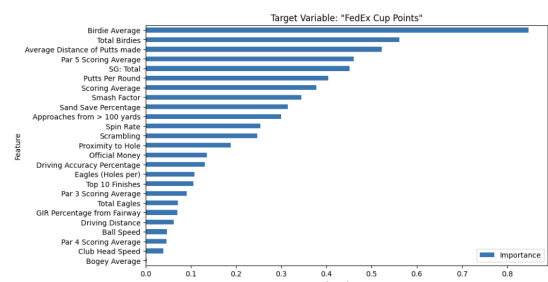
The Harkness system was not completely flawed, but it over-biased matches at special tournaments. Currently, the Professional Golfers’ Association of America does the same.² Initially, we wanted to bring an Elo-like ranking system to the golf world so that Major events, such as the Master’s or the PGA Championship, would not overly dictate who finishes the regular season with the highest FedEx Cup point total. At the same, we wanted to see if we could statistically justify why top PGA golfers were also top PGA finishers. First and foremost, golf is not a direct competition between two players. Rather, hundreds of players can partake in a tournament, but only one golfer from the field wins. If we were to use an Elo-like system that only updated a player’s ranking with their wins, many players would only lose rating points for the duration of their career. Secondly, and perhaps more subtlety, there is no game theory involved in golf even in the rare occasions it can be played head-to-head; each player simply tries to optimize to their lowest score, which is completely independent from something like chess where one player’s move directly impacts their opponent. As a result of these differences, we decided to take an alternative approach.

Although the FedEx cup point rankings might be partially flawed, they are still a useful metric of where players fall in the field. We decided that we wanted to investigate whether or not a player’s statistical background could predict where they finish in the FedEx rankings with a machine learning approach. More specifically, we wondered if we could leverage stats, like driving distance, scrambling, total number of birdies, or par 4 scoring average for example, to predict whether a player would finish in the top 10 FedEx Cup point earners in a given season.

We began addressing this problem by acknowledging that golf is one of the hardest sports to seemingly predict. For one, the field is vast; every PGA Tour event has well over 100 golfers in the field. Moreover, golf has several factors that make predicting outcomes highly complex. For example, weather can have a huge impact on a tournament. If the course is extremely windy on a given day, certain golf styles (players with lower than average shots in trajectory) will have an advantage. The opposite is also true; on a particularly still day, the golfers who hit the ball with a high trajectory have the advantage. Putting, where tournaments are won or lost, is also highly random, as players can “get hot” or “cool down” with their ability within a few holes. Given both previous results and raw statistics, predicting golf outcomes is a highly promising problem to explore despite the stochastic sport.

II. MODEL DESCRIPTION

Via Kaggle, we found a dataset that detailed key stats for every PGA golfer who competed from 2015-2019. The data set was comprehensive, with over 29 columns of data for each player in each year played on the PGA Tour. There were several outcome variables, and we wanted to see how they interacted with each other. We initially focused on FedEx Cup Points as our main outcome variable and used logistic regression to assess the importance of other factors, which we believed to be the best measure of success.



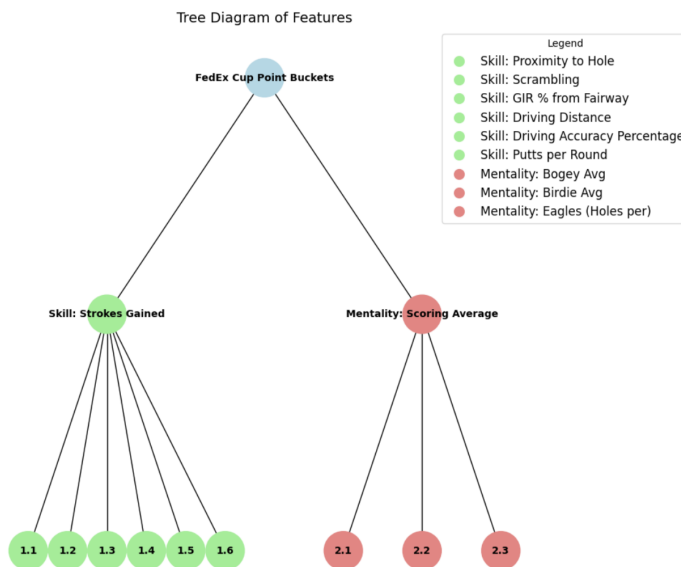
¹https://en.wikipedia.org/wiki/Elo_rating_system

²https://en.wikipedia.org/wiki/List_of_point_distributions_of_the_FedEx_Cup

We did not choose the other two outcome variables, Top 10 Finishes or Total Money earned for good reason. For Total Money earned, we see a massive disparity in the total purse (money available to win) at Major tournaments as compared to regular weekend events. We felt that total money would therefore not be the best indicator of season long success, as the extra weight on Major tournaments would in fact lead to skew. For Top 10 finishes, recognized that some of the bottom players do not finish in the top 10 at all, and the very top players finish in the top 10 in nearly half of the tournaments that they play in. Therefore, we thought this discrete distribution would not be ideal to test as our outcome variable.³

With our target variable in place, we set out to identify how a golfer's skills lead to their success. At a high level, we sought to quantify two important factors to PGA Tour success: innate golf ability and mental edge. Golf is considered to be "90%" mental by many that play the game (our data does not necessarily agree), so we committed to proving or disproving the truth behind the claim.

Our decision was to measure innate golfing ability through the Stroke Gained metric, and the mental side (controlling for strokes gained, the actual golfing ability) through scoring average. We will elaborate on the features used for our modified strokes gained and scoring average metrics in the subsequent sections:

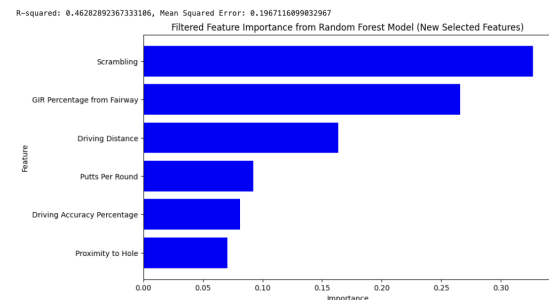


A. Strokes Gained

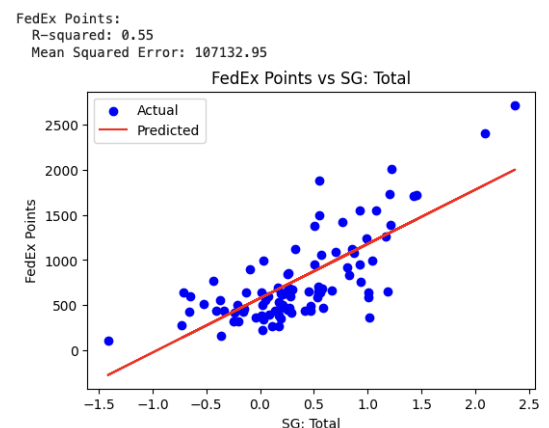
Strokes gained, which measures how many strokes the golfer improves over par in a given round, provides clear insight into a golfer's talent because it quantifies performance relative to the competition, accounting for the difficulty of the course and the average performance of other players. A more comprehensive breakdown of how total strokes gained is calculated is more of a sum of the parts walk. The algorithm measures strokes gained across all the primary parts of the

game of golf, such as driving, iron/approach shots, putting (while on the green), scrambling (how well you recover if you miss the green), and more. The sum of all of these components leads to a Strokes Gained (SG): Total category. SG allows us to assess a golfer's skills in various contexts, rather than relying solely on raw scores, which can be influenced by external factors such as course conditions and player strategy. By focusing on strokes gained, we could more accurately identify exceptional talent and determine which golfers are consistently outperforming their peers. Starting with "SG: Total" as the target variable, we generated the following feature importance plot with a random forest regression model with a collection of decision trees and 80% test-train split. A decision tree is a model that makes predictions by splitting data into branches based on feature values. Each node represents a feature, while leaves represent the predicted outcomes. A random forest regressor combines multiple decision trees, averaging their predictions to improve accuracy and reduce overfitting. More mechanically, we introduce randomness in tree selection and feature usage, which enhances model robustness and generalization, making it effective for our PGA Golfer Statistic dataset.

Iterating on this idea, we saw that many of these features were heavily correlated. For example, club head speed is a near perfect predictor of driving distance. Therefore, we decided to include only the most "important" of these metrics that contribute to the following golf attributes:



Like we suspected, there was a strong correlation between strokes gained and FedEx Cup points:



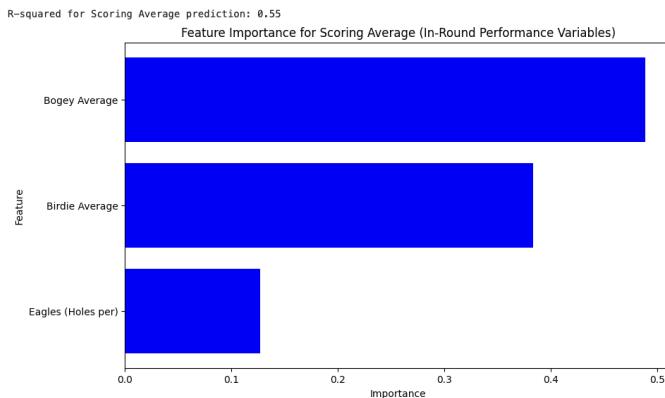
³<https://www.kaggle.com/datasets/jychoi87/pga-tour-top-200-player-data-20152019/code>

B. Scoring Average

After our work with predicting Strokes Gained, we turned our attention to scoring average and tapped features in our dataset tied to overall round performance. Again, we did this by controlling for the golfer's innate ability captured already within the Strokes Gained metric. We once again ran a Random Forest Regression model and found that Par 4 average was by far the most important.

At first, this seemed an extremely surprising result. After all, if you look at dominant runs in golf (i.e. Tiger Woods in the early 2000s, Scottie Scheffler recently), dominating Par 5's was the key driver to their success. Anecdotally, this is also true; all golf analysts stress that success at the PGA level comes from making birdie or better on as many of the Par 5s as possible. Removing these implicit biases, we see that our result reflects the number of weighted number of holes on the course: an average 18-hole golf course, there are 10 par 4s, 4 par 5s, and 4 par 3s. Given that par 4s encompass a larger percentage of the total hole count than par 3s or 5s (roughly $\frac{2}{3}$ of all holes), the extra weight on par 4 average was reflected in the feature importance plot.

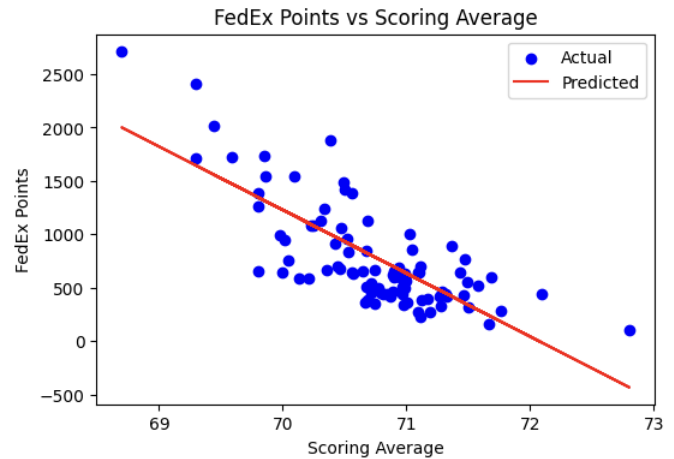
Because of this result and massive weight on Par 4 scoring average, we redirected our attention to Bogey Average, Birdie Average, and Holes per Eagle (measured by holes per as eagles are much more rare). A bogey in golf means that a player took 1 more stroke than expected to complete a hole (an unfavorable result). A birdie is the opposite: it took the player 1 stroke less than expected to finish. An eagle, which almost exclusively happens on Par 5s, occurs when it takes 2 strokes less to finish. Running the model on scoring average yields the graph:



For our target variable, scoring average, we see that limiting mistakes (bogeys) is even more important than capturing upside (birdies). Eagles, by in large, are very rare and normally occur once every few rounds. This ties very heavily that scoring average is driven by mental fortitude, and limiting mistakes.

Again, as expected, we see that scoring average has a strong correlation with FedEx cup points.

FedEx Points:
R-squared: 0.58
Mean Squared Error: 101581.61



C. Building the Model

We have established two principles:

1. **SG: Total** is a measure of how good a golfer is, which is driven by how good a golfer primarily is at capitalizing on opportunity.

SG: Total = $.33 * \text{Scrambling} + .27 * \text{GIR from Fairway} + .16 * \text{Driving Distance} + .09 * \text{Putts per Round} + .08 * \text{Driving Accuracy} + .07 * \text{Proximity to Hole}$

2. **Scoring Average** is a measure of how strong a golfer's mental fortitude is; the talent on the PGA Tour level is so strong that the difference in limiting mistakes, which we found to be the key driver here, is not a difference in skill but one of differing mental approaches.

Scoring Average = $.49 * \text{Bogey Average} + .38 * \text{Birdie Average} + .13 * \text{Eagles}$

In order to fit the desired scope to our model, we decided to add a cutoff point to necessitate that all of the players that we were considering were PGA Tour regulars. That is, a golfer played the majority of the season. This allowed us to exclude any players that only played in a select few tournaments. These players, of course, may have performed exceptionally well in these tournaments, but their small participation over the whole season would still mean that they would not have placed in high in the season long FedEx rankings. To restrict our analysis to regular players, we trained and tested on players that recorded > 200 birdies over the course of the season. This is almost entirely non-restrictive to players that played a full season, no matter how good or bad they were. It also served its function of eliminating those with a few number of tournaments played.

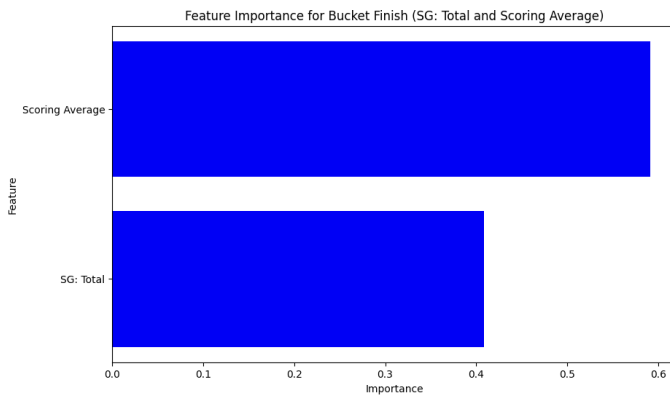
We built our prediction model to be a combination of the relative weights of our SG: Total and Scoring Average metrics. First, we devised a rating system for each of the

Strokes Gained and Scoring Average attributes using our feature importance results.

For Strokes Gained, we transformed the importance factors into weights assigned to the normalized statistics for each of the following: Scrambling, GIR Percentage from Fairway, Driving Distance, Putts per Round, Proximity to Hole, and Driving Accuracy Percentage. This results in a predictive SG rating for each player.

For scoring average, we took a very similar approach. We again used the importance factors as weights to assign to the normalized Bogey Average, Birdie Average, and Eagles (Holes per). One caveat here is that unlike the other statistics, the more bogeys that you had the worse it was for your scoring average. Therefore, we took care to account for this in our model when normalizing.

Finally, we now have each golfer's (each season treated independently) relative predictive SG and scoring average. Again, we now have a relative rank for the golfer's innate golf ability and their mental fortitude. To decide how to scale these ratings, we ran a simple random forest regression on SG: Total and Scoring Average from our original dataset. That said, we tested on a binary variable that we cast into our data that detailed whether a player finished in the top 10 of the FedEx Cup Point standings. We called this variable "Top 10 Finish," which is a season-long metric.⁴ We then used these factors to assign the proper weight to each player's relative mental fortitude and golf ability.



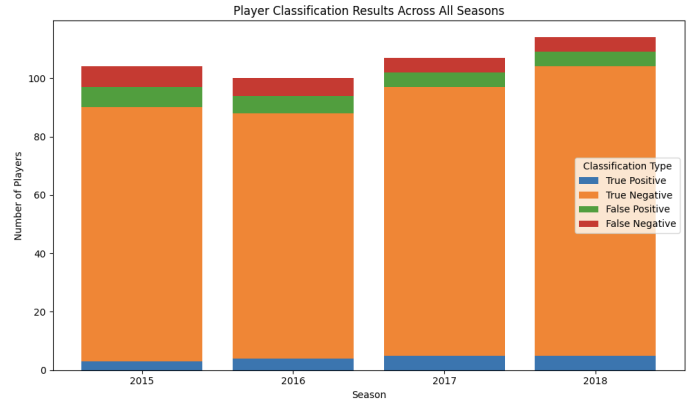
Note: We were skeptical of the initial weights placed upon Scoring Average and SG: Total due to a very low R^2 of 0.19 for bucket prediction, so we decided to sensitize them in our final model.

$$\text{Power Ranking} = .59 * \text{Scoring Average} + .41 * \text{SG: Total} \quad (1)$$

We have now cemented the foundation of our model. On the 2015-2018 train set, we have derived weights associated with a broad spectrum of skill-based and mentality-based attributes that describe a PGA golfer's ability on the course. Overall, we

⁴This variable is not to be confused with "Top 10 Finishes," which qualifies the number of times a player finished in the Top 10 of a tournament in a given season!

feel that the model is reflective of reality in terms of the test data, with the players at the top of our rating system being squarely recognized as the best golfers in the world at the time. With around 90% accuracy, our model reflected whether or not a player finished in the top 10 of FedEx Cup Point rankings in each of the train seasons:



Now, it is time to test our model on the 2019 season.

III. ANALYSIS

Testing our model on the data reaffirmed two things:

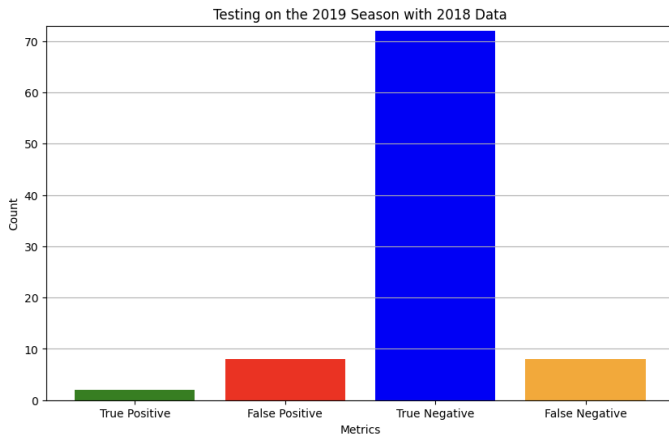
- 1) Vegas' job of setting odds is very difficult with diligent arbitrage hunters in the market.
- 2) Golf is very hard to predict.

We were rather pleased with our overall results, which we tested two different ways.

First, we wanted to see if the 2019 season was similarly behaved as the prior seasons used to train the model. To do this, we kept the weighting coefficients on SG: Total and Scoring Average that we derived using our training set. Within both those categories, we kept the same weights on each inherent statistic. We then applied each player from 2019's actual stats to get our predicted top 10, and compared this to the actual top 10 that unfolded.

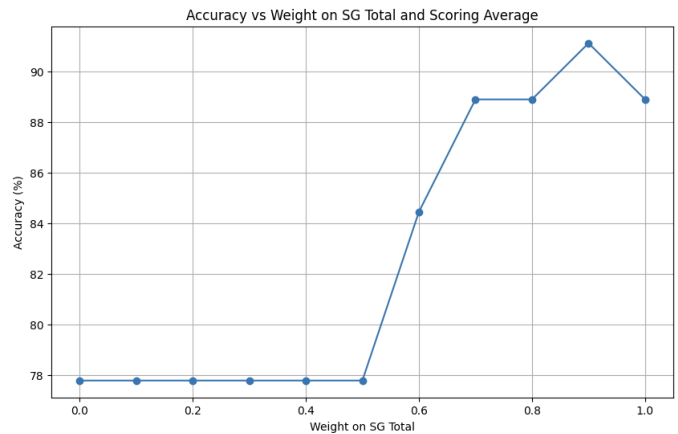
Note: we understand that this does not assess the predictive capabilities of our model from an odds perspective, as we make our predictions with data that is "from the future". Meaning: we would plug in the golfer's scrambling rate and multiply this by the appropriate coefficient (and so on), until we reach their power rating (as described in the model). Our sense check yielded results consistent enough to our training data

Then, we wanted to of course test our model's actual predictive abilities. To do this, we again utilized the 2019 season from the Kaggle set that we did not train the data on. We ran through the data set and looked for golfers that were given our power rating in 2018 and played in 2019. We then used the power ratings (again, these do not align perfectly with their actual finish position) from 2018 to predict where they would finish in the 2019 season.



At first, this seemed a disappointing result. Our model was not necessarily wrong; it just was not as accurate as we would have liked it to be. This could have been for a lot of reasons. First, the 2019 golf season may well have deviated from the behavior that our ML model was trained on using the previous seasons. However, our testing of plugging the actual 2019 statistics into the built model proves this wasn't exactly the case. Perhaps, it was because we assigned the improper weights to each of the factors that we used to build SG: Total and Scoring Average. However, we felt that these were not over-fit to the training set but fair to the impact of SG: Total and Scoring Average given our model's inherent simplicity. We realized, however, that our weight assignment to the relative SG: Total and Scoring Average ranks were imprecise, and the justification wasn't as strong as our others. This is therefore where we decided to focus most of our sensitivity analysis.

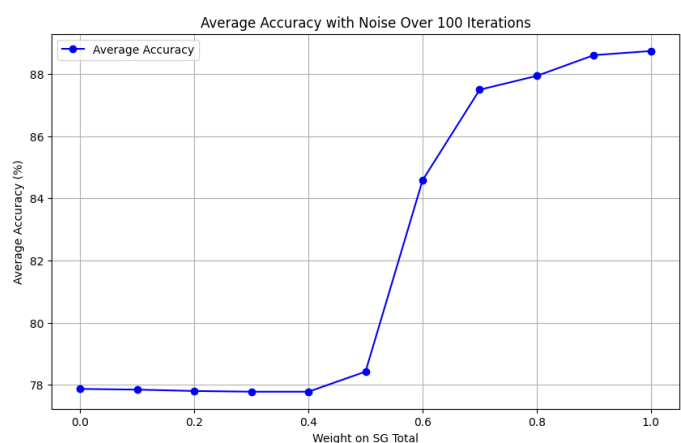
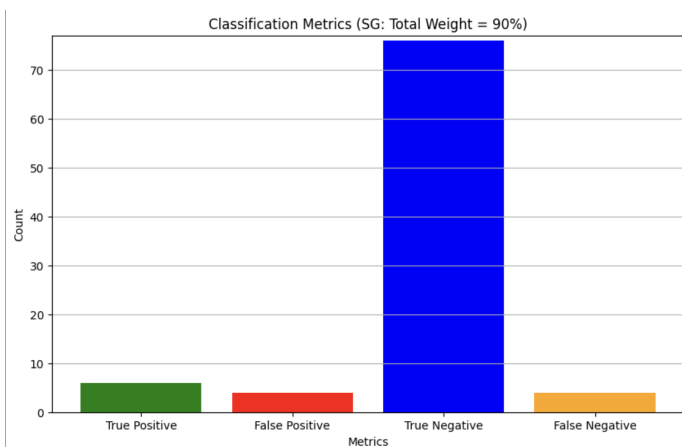
It was our sensitivity analyses that proved the most exciting. We wanted to see how our model's accuracy on the test data changed as we shifted our weights on SG: Total and on Scoring Average. To this end, we simply iterated through different weight coefficients to our linear combination formula, and graphed each respective accuracy. For our sensitivity analysis, we assumed a noiseless system on the player data from 2018:



This result was incredibly promising. We were able to see that there exists an optimal weight of coefficients to our formula, right at the hitch of the 90% SG: Total to 10% Scoring Average. This seems very counter intuitive - how is it that last season's Scoring Average is that insignificant in our model? Lots of this has to do with the high correlation between birdie make percentage and Strokes Gained. The innate attributes that we use to build out Strokes Gained, such as GIR from Fairway, Driving Distance, and Putts per Round are immediately indicative of how many birdies a play may make. Meaning: it is very likely that SG: Total captures (and maybe more effectively) the effects of Scoring Average that birdie average contributes to.

Further, we can justify SG: Total being weighted heavier because it is indeed more process, rather than outcome, focused. Process oriented approaches are often much stronger at predicting future outcomes, as it focuses more on an aggregate of controllable aspects of golf. Whereas scoring average, while it does indeed also capture some of these components, also factors in more external factors (such as weather, random pin placements, etc.). These external factors, while they have a definitive impact on current results, are far less predictive into the future.

Finally, we decided to try and see what happens when we introduce a feature of random noise to our sensitivity. This noise would attempt to add some of the randomness that the external factors described above may introduce to our model.



This graph takes a very similar shape to our first sensitivity graph. We see that, again, the optimal mix is with the 90% weight towards SG: Total and a 10% weight towards Scoring Average. The matching of these two optimal points is a powerful result - that even with an amount of randomness, our result does not deviate.

Now, we do see our model accuracy decrease relative to the model with no noise. This is entirely expected - given our model is indeed accurate in prediction, introducing unbiased randomness can only bring down this accuracy. Therefore, the confirmation of our optimal SG: Total to Scoring Average weighting, even with the drop in accuracy, shows that our model indeed works in real world settings with randomness.

IV. DISCUSSION

Our results are encouraging, but they could be viewed as misleading. In a field of 100 players, 10 must end up in the top 10 players. If every player was incorrectly classified, 90 players would still be classified as not making the top 10, which means that we would have 80% accuracy if every prediction was missed. If someone were to pick players at random, the expected value of selecting a player in the top 10 would be 1 after 10 guesses. So, to outperform complete randomness, we needed to correctly identify more than one player correctly in the top 10. We set a goal at having an accuracy of over 85%, which we attained by optimizing our SG: Total and Scoring Average weights in our sensitivity analysis. With our optimal weights, we increased our final accuracy to 89.37% in the noisy system with a 0.9 weight on our modified SG: Total metric and 0.1 weight on our Scoring Average metric. Ironically, this would imply that the game is only “10%” mental and “90%” physical: the utter opposite of the age-old adage. While we beat our initial accuracy projection, there are several places where we felt our model could have been improved.

- 1) **Past FedEx Cup Rankings:** If we had used scaled FedEx Cup Rankings from the k th year to motivate the $(k + 1)$ th year, we might have been able to control for consist top performance over time.
- 2) **Feature Selection:** We selected features from our importance plots, but by bifurcating our model into SG: Total and Scoring Average and finding each sub-feature’s importance to each parent metric, we might have oversimplified or overlooked other pertinent statistics.
- 3) **Randomness:** We approximated randomness in our simulated 2019 data by adding a coefficient with mean 0 and a standard deviation determined by the span of the Power Rankings data we derived to each Power Ranking. This was an elementary way of simulating noise. A more rigorous way of investigating the impact of noise would have been to add it to each feature in the training data instead of our our test set. This might have influenced the Power Rankings in a more profound way.
- 4) **Momentum:** We had no way of quantifying golfer momentum. As has been mentioned previously in the paper, Scottie Scheffler is on a historic run right now, but our model would only take into account a subset of

his statistics in a present-day snapshot. To account for momentum, we would search for a separate dataset that quantifies Top 10 tournament finishes by date. We would scale each player’s power ranking with a small epsilon (added to one) that would scale with consecutive wins.

- 5) **Dataset:** Our dataset was missing several columns that we felt would have been critical for our analysis. There were no metrics on how often a player attained par or worse than bogey on a hole. These metrics would have impacted our Scoring Average metric heavily. Moreover, we lacked a history of events players had played in and any composition of how a player’s FedEx points had been earned. Finally, we lacked how many rounds a player had played in a year. If we had known, we would have removed our 200 birdies condition and implemented a minimum bar for number of rounds played.
- 6) **Correlated Factors:** We recognize that many of the factors in our model are heavily correlated with each other. This is part of what makes golf so hard and fun to predict. It is nearly impossible to perfectly separate out elements from each other to produce a predictive model, but we controlled for this where possible.

V. SUMMARY

Motivated by the invention of the Elo Ranking system in chess, we set out to create a skill-based power ranking system for golf that could simulate an upcoming year’s Top 10 FedEx Cup point earners with data from past seasons. We focused on factors we categorized as mental and physical, and by leveraging a Random Forest Regressor, we derived each factor’s relative importance to its parent statistic. By running a sensitivity analysis on the parent factors, SG: Total and Scoring Average, we demonstrated that by analyzing the statistical background of PGA players from 2015 – 2019, we could predict the top 10 finishers in the 2019 season with up to 89.37% accuracy. Our model has its limitations, and we acknowledged how our dataset, feature selection methodology, correlated factors, randomness simulation, and failure to account for momentum could have been further investigated to improve our broader model. In the future, we would like to increase the accuracy calculation within our model by working with a larger dataset and attempting to rectify the issues mentioned above. Furthermore, we would like to get the betting odds via an API from a major casino and compare our own prediction against that of Vegas. We look forward to pursuing this project in greater depth in the future, and we are confident that we have developed a strong launchpad for future inquiry.

APPENDIX A PYTHON NOTEBOOK

[Click here](#) to access our Python supplement. We have cultivated a notebook that mirrors the progression of our paper and justifies our graphics.

ACKNOWLEDGMENT

Thank you to Professor Brenner, TF Elle Weeks, enhanced CA Erik Wang, and our respective sections for their feedback and guidance throughout the development of this project.

We would like to give a necessary attribution to ChatGPT, which helped us in random spots in our notebook. We derived all of the logic and initial code ourselves, but ChatGPT lent us a hand with as we sought to create powerful visuals and debug persistent syntax errors.

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